



MRS6130-P1806

mmWave Radar Module

User Guide

Number of Pages: 17

Document Version: 1.3

Release Data: 2025-06-27



Revision History

Version	Version Description	Date
1.3	Add packaging information Correct the description of the pin	2025-06-27
1.2	Format adjustment	2025-02-14
1.1	Modify the voltage description	2024-07-01
1.0	Initial release	2024-03-15

Table of Contents

Revision History.....	ii
Table of Contents.....	iii
List of Figures.....	iv
Table of Contents.....	v
1 Overview.....	1
1.1 Module introduction.....	1
1.2 Module specifications.....	1
1.3 Application scenarios.....	2
2 System Application Block Diagram.....	3
3 Pin Definition.....	4
3.1 Pin layout.....	4
3.2 Pin description.....	4
4 Peripheral.....	6
4.1 GPIO.....	6
4.2 UART.....	6
4.3 PWM.....	6
5 Hardware Usage.....	7
5.1 Hardware start up.....	7
5.2 Firmware burning.....	7
5.3 Demo interface.....	7
6 Electrical Characteristics.....	8
6.1 Power supply operating conditions.....	8
7 Peripheral Design Schematic Diagram.....	9
8 Module Size and PCB Packaging.....	10
8.1 Module dimensions.....	10
8.2 Recommended PCB packaging.....	10
9 Packaging Information.....	11
9.1 Carrier tape size (Unit: mm).....	11
9.2 Reel size: D=38cm 2750 pieces/reel.....	11

List of Figures

Figure 2-1	System Application Block Diagram.....	3
Figure 3-1	Pin Layout.....	4
Figure 7-1	Schematic Diagram of Peripheral Design.....	9
Figure 8-1	Dimensions Of Mrs6130-P1806 Module.....	10
Figure 8-2	Recommended PCB Package.....	10
Figure 9-1	Tape Carrier Size.....	11
Figure 9-2	Reel Diagram.....	11

Table of Contents

Table 1-1	Module Specifications.....	1
Table 3-1	Description of Module PIN Functions.....	4
Table 6-1	Operating Conditions of Module Power Supply.....	8

1 Overview

1.1 Module introduction

MRS6130 is a module developed based on the new generation of ultra-low power single-chip 60 GHz 1T3R FMCW AiP radar sensor chip RS6130 from Possumic. The module integrates a 1T3R AiP radar chip and flash, enabling rapid application development for developers after direct power supply to the module. The module includes four low-power management modes: shutdown, standby, idle, and working. It not only features low power consumption in standby mode but also in sensing through our fast burst engine and high-performance RF transceiver. Its configuration and data acquisition are implemented through the UART interface. The module is divided into three subsystems:

- **mmWave RF and BB Subsystem:** This module encompasses all high-performance radar RF/analog and base band digital signal circuits. It also incorporates a linear frequency modulated pulse generator and a Tx-to-Rx state machine.
- **Processing and Application Subsystem:** It comprises a 32-bit RISC CPU and a Radar Hardware Accelerator (HWA) for diverse application processing and task management. Our proprietary HWA IP is utilized for handling specific radar computations (such as FFT, CFAR, etc.), compression, as well as moving target perception and localization.
- **Power and clock subsystem:** RS6130 is a complex single-chip SoC with multiple functional systems. We utilize the on-chip integrated PMU system and clock subsystem for centralized management.

1.2 Module specifications

Table 1-1 Module Specifications

Specifications	Minimum	Typical	Maximum	Unit	Remarks
Radar Specifications					
Operating Frequency Band	58.0	-	64.0	GHz	-
Continuous Frequency Scanning Bandwidth	-	6	-	GHz	-
Number of Transmit and Receive Channels	1T3R			-	It can achieve different mode configurations such as 1T1R, 1T2R, and 1T3R
Equivalent Isotropic Radiation Power (EIRP)	-	17.5	-	dBm	-
Horizontal FOV	-	120	-	Degree	-
Vertical FOV	-	120	-	Degree	-
Farthest Detection Distance	-	-	20	m	It depends on the size of the RCS of the detection target
Recent Detection Distance	-	-	0.01	m	Take human targets as a reference
Hardware Specifications					
Module Dimensions	-	18.5x6.5	-	mm ²	-
VCC Supply Voltage	1.7	3.3	5.5	V	-
VDDIO Supply Voltage	1.7	3.3	3.6	V	-

Operating Junction Temperature	-40	-	105	°C	-
Moisture Sensitivity Level	MSL-3				-
Environmental Standards	Complies with RoHS standards				EU 2011/65/EU

1.3 Application scenarios

The MRS6130 module can be widely applied in typical AIOT intelligent perception scenarios represented by smart home appliances, intelligent security, smart health care, and intelligent cockpits, achieving high-precision and high-sensitivity detection of human body and object movements, micro-motions, breathing, and heartbeat:

- Human movement/presence detection
- Human body tracking and counting
- Human respiratory and heartbeat detection
- Gesture recognition
- Video doorbell
- Network Cameras
- Air conditioner
- Refrigerator
- Television

2 System Application Block Diagram

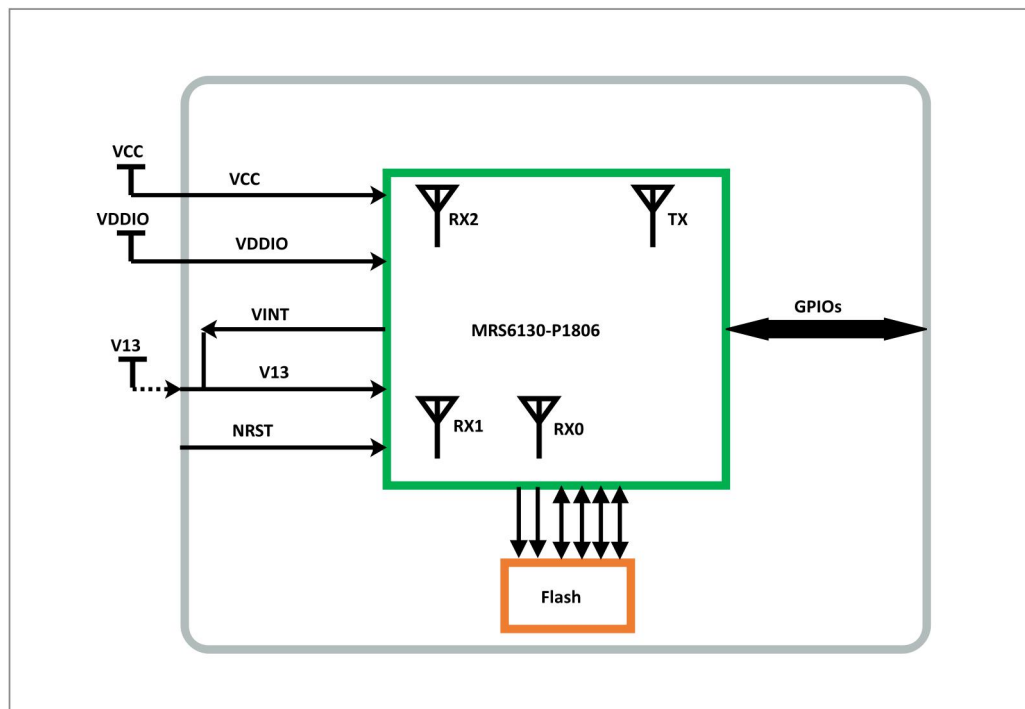


Figure 2-1 System Application Block Diagram

The periphery of the MRS6130 millimeter-wave radar module includes a power supply and four GPIO. VCC is the main power supply for the module, while VDDIO is the power supply for the GPIO. When the power supply voltage does not exceed 3.6V, VCC and VDDIO can be directly short-circuited during application. V13 is powered by the internal VINT by default.

3 Pin Definition

3.1 Pin layout

The layout of the module's PIN pins is shown in the figure below.

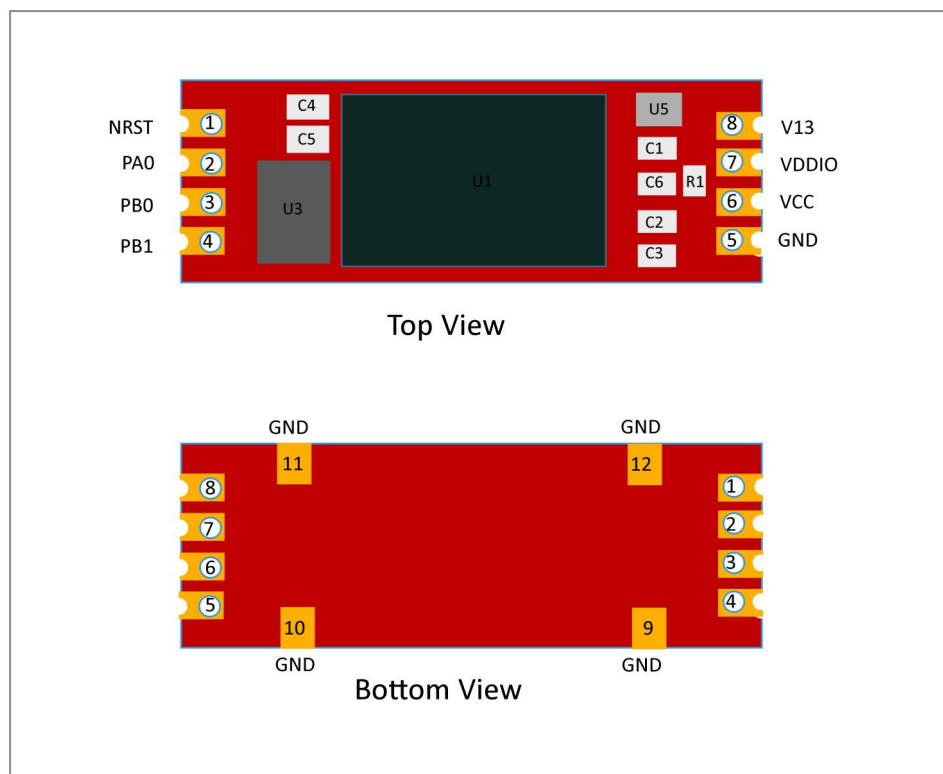


Figure 3-1 Pin Layout

3.2 Pin description

The module has 12 pins, and the function descriptions of the pins are as shown in Table 3-1:

Table 3-1 Description of Module PIN Functions

Pin Name	Serial Number	Type	Functional Description
GND	5,9,10,11,12	G	Ground Pin, Short directly to board ground plane.
NRST	1	GPIO	Module reset
PA0	2	GPIO/WUP	WIO0,UART1-TX,PWM0,GPADC7
PB0	3	GPIO/WUP	WIO16,UART0-TX
PB1	4	GPIO/WUP	WIO17,UART0-RX
VCC	6	PI	Module power supply: 1.7V~5.5V, default input: 3.3V
VDDIO	7	PI	The module IO power supply is 1.7V~3.6V, with a default input of 3.3V
V13	8	PI	The analog power supply operates within a range of 1.2V to

			2.5V, with a default output of 1.5V. It is powered by the internal VINT by default, but can also be powered by an external DCDC
--	--	--	---

Note:

- 1) PI stands for input power;
- 2) PO stands for output power;
- 3) GPIO stands for General-Purpose Digital Input/Output;
- 4) WUP represents the digital pin that can maintain the state or wake up the system in standby mode, where WIOx refers to the sequential number of the WUP type IO

4 Peripheral

4.1 GPIO

The MRS6130-P1806 module provides three general-purpose GPIO, each GPIO pin features multiple switchable alternate functions. GPIO configuration features:

- Input, output, and high impedance modes
- Input configurable pull-up and pull-down
- The output drive capability is set to 20mA by default, with one setting at 20mA and a maximum configuration of 80mA
- The IO type supports push-pull and open-drain mode configurations
- Supports 5 types of external IO interrupts: rising edge, falling edge, both rising and falling edges, high level, and low level

In low-power scenarios, all GPIO support triggering the system wake-up through interrupt mode.

4.2 UART

The MRS6130-P1806 module supports UART0 TX/RX and UART1 TX. UART is compatible with the 16550 industry standard and supports baud rates up to 3 Mbps. UART features include:

- Programmable transmit and receive baud rate
- Baud rate self-checking function
- Supports 64-byte receive and transmit FIFO
- Supports data lengths of 5/6/7/8 bits
- Support 1/1.5/2 stop bits
- Support parity check
- Supports DMA high-speed data communication
- Support UART wake-up function

UART RX has a function to detect rising edge changes and trigger an interrupt, with the number of rising edges to be detected configurable from 1 to 8. In low-power scenarios, UART RX wakes up the system upon receiving a matching rising edge.

4.3 PWM

The PWM controller can be used to output modulated digital waveform signals, supporting an adjustable duty cycle from 0 to 100%.

5 Hardware Usage

5.1 Hardware start up

Module VCC is powered by 3.3V, VDDIO is powered by 3.3V, and V13 is internally powered by VINT at 1.5V by default.

5.2 Firmware burning

Power on the module, pull PB0 pin low, pull NRST low and then high, start the hardware, enter UART0 upgrade mode (PB0/PB1), and then release PB0. Use the BurnTool to download firmware and Flash on the module.

5.3 Demo interface

The U4 interface supports docking with the MRS6130-DEV-V1 Demo development board.

6 Electrical Characteristics

6.1 Power supply operating conditions

Table 6-1 Operating Conditions of Module Power Supply

Name	Parameter Description	Minimum	Typical Value	Maximum	Unit
VCC	Module power supply input voltage	1.7	3.3	5.5	V
VDDIO	Module IO input voltage	1.7	3.3	3.6	V
V13	Internal power supply output voltage of the module	1.2	1.5	2.5	V
Ivcc	The current provided by the external power supply	-	-	500	mA

7 Peripheral Design Schematic Diagram

The application schematic diagram for connecting peripheral devices of the module is shown in the figure below.

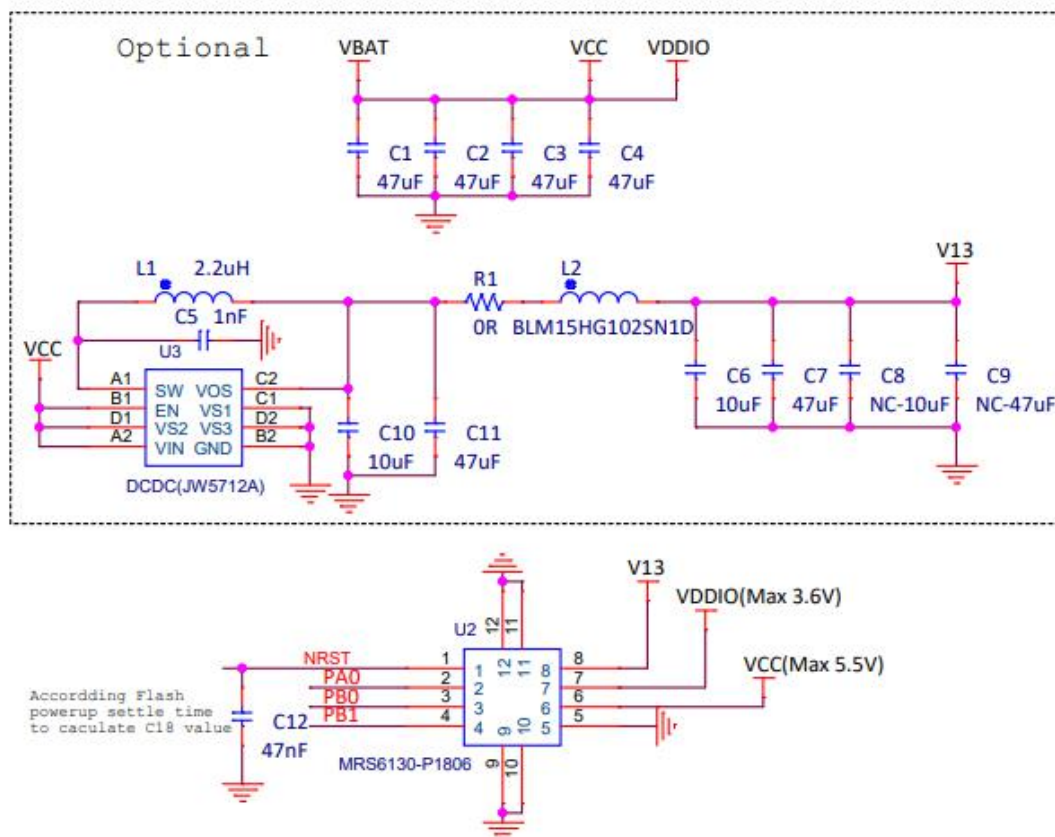


Figure 7-1 Schematic Diagram of Peripheral Design

For ultra-low power consumption scenarios, the V13 application can choose to be powered by an external low-power DCDC (with a recommended peak supply current of >500mA). By configuring the INT LDO that outputs the VINT voltage internally within the chip to be lower than the external 1.3V voltage (a recommended configuration of 1.1V), or by turning off the INT LDO after the chip is powered on, the chip can achieve automatic power supply switching.

For low-power DCDC, we recommend a low-leakage DCDC power chip for No-Loading applications. V13 is sensitive to power supply noise, so it is recommended to add an LC filter to remove power supply interference. In applications involving high-internal-resistance batteries such as coin cells, under acceptable cost conditions, the VBAT capacitor and V13 capacitor should be maximized (at least 200uF) to ensure that the maximum transient power supply ripple does not exceed 150mV.

Due to the slow power-on stabilization time of Flash, the timing relationship between NRST release and VCC and VDDIO power-on must be considered during peripheral circuit design:

- 1) When the peripheral circuit master controls NRST, after the radar module's VCC and VDDIO are powered on, NRST must be released only after the flash has stabilized
- 2) When the peripheral circuit master does not control NRST, the peripheral circuit must add a capacitor to NRST, and the size of the capacitor depends on the flash power-on stabilization time

8 Module Size and PCB Packaging

8.1 Module dimensions

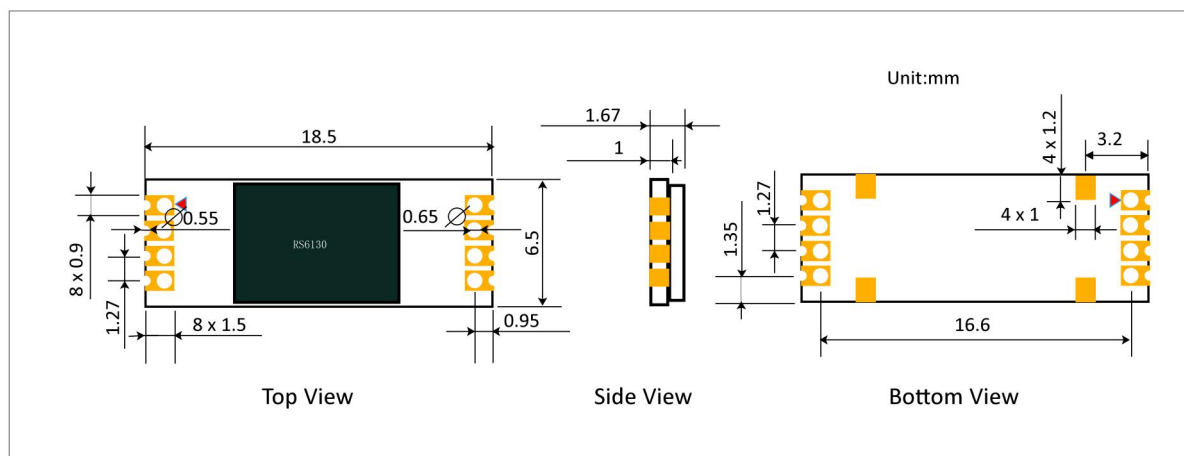


Figure 8-1 Dimensions Of Mrs6130-P1806 Module

8.2 Recommended PCB packaging

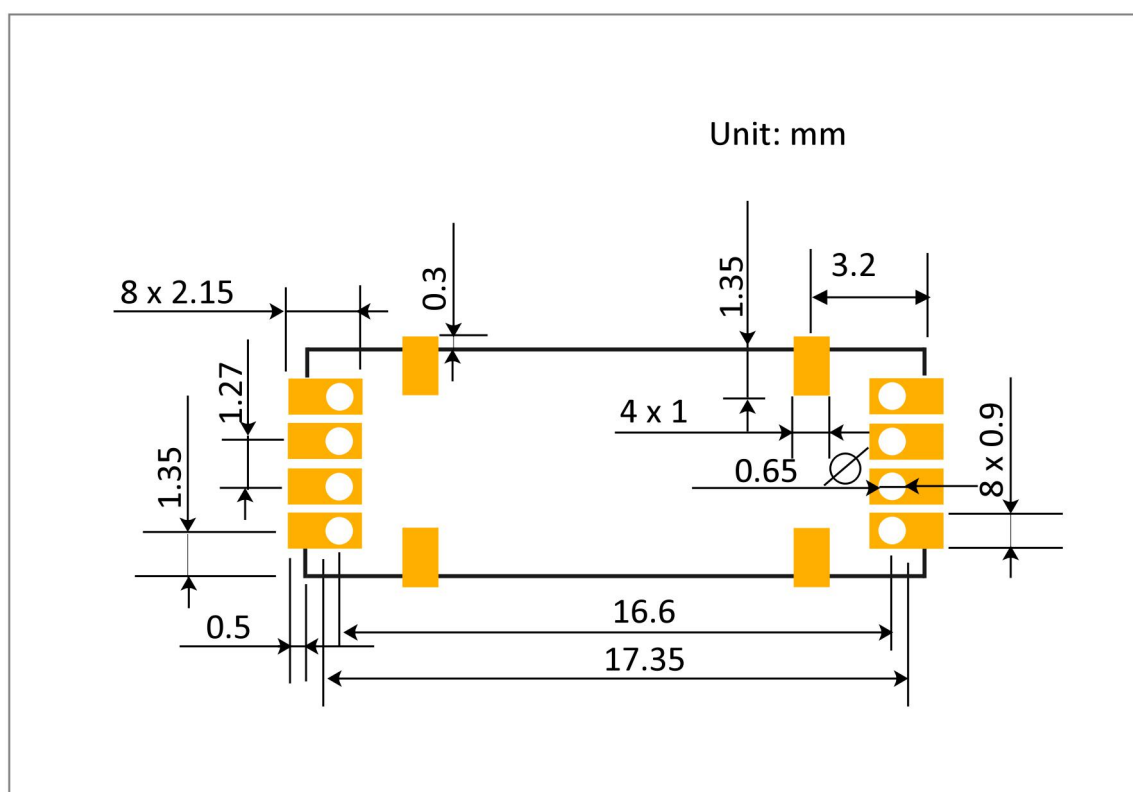
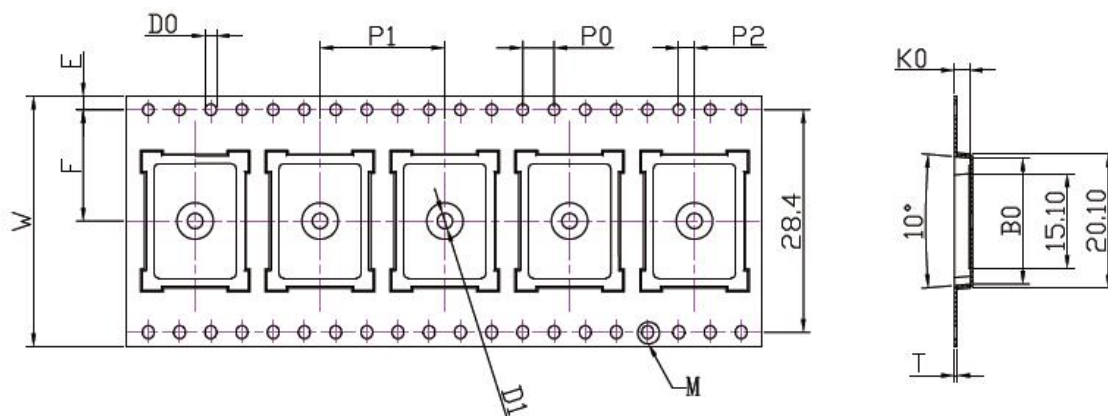


Figure 8-2 Recommended PCB Package

9 Packaging Information

9.1 Carrier tape size (Unit: mm)



Symbol	A0	B0	K0	P0	P1	P2
Specs	6.95 ± 0.1	19.10 ± 0.1	2.20 ± 0.1	4.0 ± 0.1	12.0 ± 0.1	2.0 ± 0.1
Symbol	W	T	E	F	D0	D1
Specs	32.0 ± 0.3	0.3 ± 0.05	1.75 ± 0.1	14.2 ± 0.1	$\phi 1.5^{+0.1}_{-0}$	$\phi 2.0^{+0.1}_{-0}$

Figure 9-1 Tape Carrier Size

9.2 Reel size: D=38cm 2750 pieces/reel

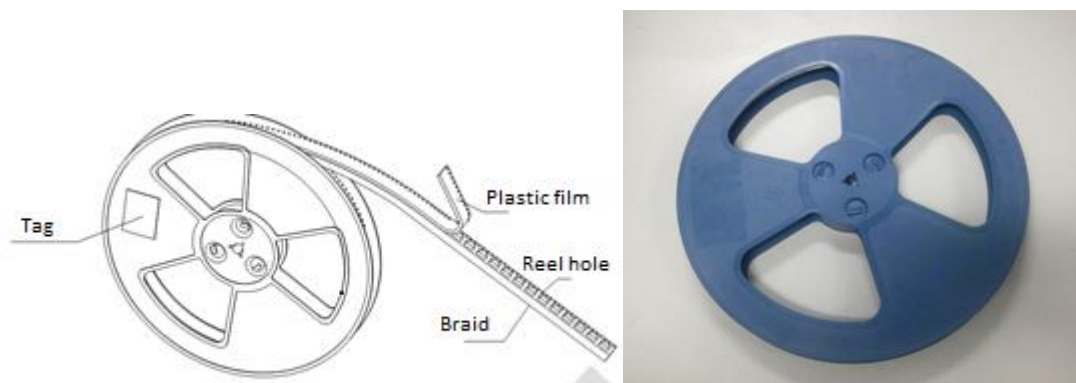


Figure 9-2 Reel Diagram

Disclaimer and Copyright Notice

The information provided by our company in this document is accurate to the best of our knowledge, but we do not guarantee that the content is completely error-free. All liability for any damages arising from the use of this document (including but not limited to indirect, incidental, or special losses) or any improper use resulting from non-compliance with the instructions herein (including but not limited to over-voltage, over-frequency, or over-temperature operations) is explicitly disclaimed.

The content in this document, including URL references, may differ from the actual content due to product version upgrades or other reasons, and is subject to change without notice.

THIS DOCUMENT IS PROVIDED "AS IS" WITH NO WARRANTIES. NO WARRANTY IS PROVIDED TO THIS DOCUMENT FOR ITS MERCHANTABILITY, NON-INFRINGEMENT, FITNESS FOR ANY PARTICULAR PURPOSE, NOR DOES ANY WARRANTY OTHERWISE ARISING OUT OF ANY PROPOSAL, SPECIFICATION OR SAMPLE.

All liability, including liability for infringement of any patent rights, relating to use of information in this document is disclaimed. No licenses express or implied, by estoppel or otherwise, to any intellectual property rights are granted herein.

Copyright ©2026 Possumic Technology Co., Ltd. All rights reserved.